Algebra: equations and problem solving

Middle school - Year 4 (4AM) | Lesson + examples + exercises + answers

How to use this sheet

Read the lesson, try each exercise without the answer key, then correct your work. These original exercises are for practice and should be checked against your teacher instructions.

1. Key idea

An equation says two expressions have the same value. To keep it balanced, perform the same operation on both sides.

Example: x + 7 = 19. Subtract 7: x = 12. Check: 12 + 7 = 19.

2. Worked examples

3x = 21, so x = 7. Divide both sides by 3.

2x + 5 = 17: subtract 5 to get 2x = 12, then divide by 2. So x = 6.

4x - 3 = 2x + 9: subtract 2x, add 3, then divide by 2. So x = 6.

3. Practice

1) x + 9 = 23. 2) 5x = 45. 3) 3x - 4 = 20. 4) 2x + 7 = 19.

5) 5x - 6 = 2x + 12. 6) Four times a number plus 3 equals 31. Find it.

7) Two consecutive integers add up to 41. Find them.

4. Answer key

1) x=14. 2) x=9. 3) x=8. 4) x=6. 5) x=6.

6) 4x+3=31, so x=7. 7) n+(n+1)=41, so n=20; integers are 20 and 21.

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